

Independent equal-length paths: an earlier theta-graph theorem

Literature-already-answered note for arXiv:0102211

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Status. **Question 5.4 has an affirmative answer in earlier literature.** The requested configuration is exactly a balanced theta graph. Theorem 1 of Faudree–Simonovits, *On a Class of Degenerate Extremal Graph Problems*, *Combinatorica* **3** (1983), proves the required extremal upper bound. Curiously, the answer predates the 2001 source paper.

The source question

The source calls two paths with the same endpoints *independent* when they share only those endpoints. For fixed integers $s, l \geq 2$, Question 5.4 asks whether there is a constant D such that every graph on v vertices and more than

$$Dv^{1+1/s}$$

edges contains l independent paths of length s with the same endpoints.

Two paths with equal endvertices are called independent if they have only their endvertices in common.

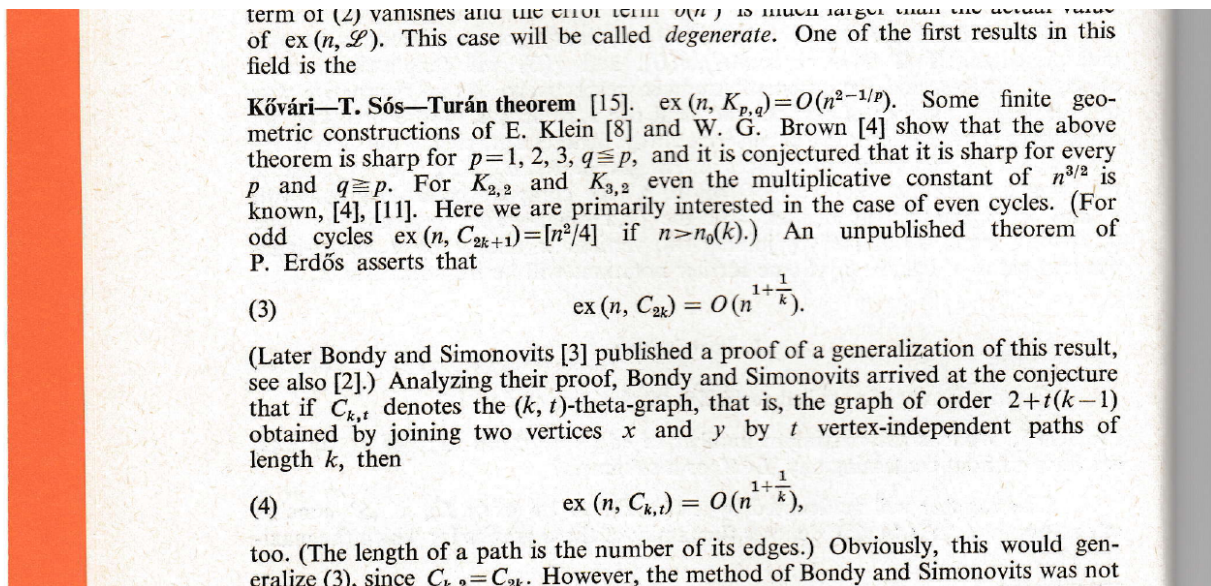
Question 5.4 Let G be a graph on v vertices with e edges. Let $s, l \geq 2$ be integers. Is it so that there is a constant D such that if $e > Dv^{1+1/s}$, then G contains l pairwise independent paths of length s with equal endvertices ?

Source evidence: arXiv:0102211, PDF page 11 (unaltered crop).

Exact identification with a theta graph

Let $\Theta_{s,l}$ be the graph formed by two vertices joined by l internally vertex-disjoint paths, each having exactly s edges. The source's word "independent" is precisely "internally vertex-disjoint," so its requested subgraph is $\Theta_{s,l}$.

Faudree–Simonovits write this graph as $C_{k,t}$ and define it in exactly these terms:

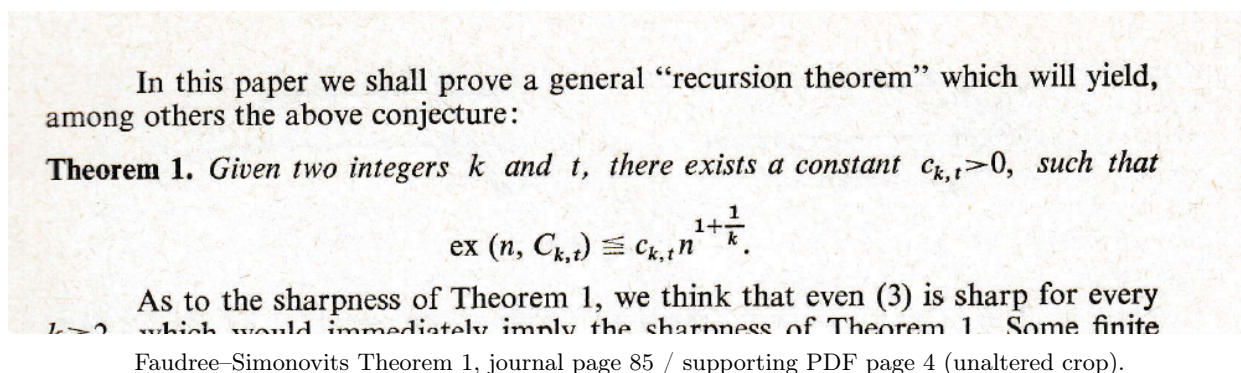


Answer-source evidence: Faudree–Simonovits (1983), journal page 84 / supporting PDF page 3 (unaltered crop).

The theorem and the one-line deduction

For every fixed pair of integers k, t , Faudree–Simonovits prove that there is a constant $c_{k,t} > 0$ such that

$$\text{ex}(n, C_{k,t}) \leq c_{k,t} n^{1+1/k}.$$



Set $k = s, t = l$, and $D = c_{s,l}$. If a v -vertex graph G had more than $Dv^{1+1/s}$ edges but contained no l independent length- s paths with common endpoints, then G would be $C_{s,l}$ -free and hence

$$e(G) \leq \text{ex}(v, C_{s,l}) \leq c_{s,l} v^{1+1/s} = Dv^{1+1/s},$$

a contradiction. Thus the answer to Question 5.4 is *yes*, with a constant depending only on s and l .

Quantifier and terminology audit

Feature	Question 5.4	Faudree–Simonovits Theorem 1
Fixed parameters	$s, l \geq 2$	k, t fixed
Forbidden graph	l internally disjoint s -paths	$C_{k,t}$, the same theta graph
Edge threshold	$Dv^{1+1/s}$	$c_{k,t}n^{1+1/k}$
Conclusion	a copy exists above threshold	contrapositive of the extremal bound

No induced-subgraph condition is present in either statement, and both use path length to mean number of edges. “Vertex-independent” in the 1983 paper and “independent” in the source both permit the two common endpoints and forbid all other intersections.

Chronology and novelty status. The answering theorem appeared in 1983, eighteen years before arXiv:0102211. This packet records a missed earlier-literature resolution; it does not claim a new theorem or a new proof.

Human review. Check Question 5.4 on source PDF page 11, the definition of $C_{k,t}$ on supporting PDF page 3, and Theorem 1 on supporting PDF page 4. The entire logical transfer is the substitution $(k, t, n) = (s, l, v)$ followed by contraposition.

References

- [1] S. Neuwirth, *Lacunary matrices*, arXiv:math/0102211 (2001), Question 5.4.
- [2] R. J. Faudree and M. Simonovits, *On a Class of Degenerate Extremal Graph Problems*, *Combinatorica* **3** (1983), no. 1, 83–93, Theorem 1.